

CS 6501 Week 1

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CS 6501 — Week 1 Survival Guide

What is Multi-Robot Navigation (MRN)?

Multi-Robot Navigation studies how multiple robots make decisions to move safely and efficiently in shared environments while interacting with each other under uncertainty.

Core Modeling Idea

MRN can be modeled as a **Partially Observable Stochastic Game (POSG)**.

POSG Components

- **k (Agents):** Number of robots in the system.
- **X (State Space):** True state of each robot and the environment (e.g., position, velocity, hidden intent).
- **U (Action Space):** Actions available to each robot (e.g., velocity, acceleration).
- **T (Transition Function):** How actions change the state (dynamics or simplified motion models).
- **J (Cost / Objective):** Measures performance (time to goal, safety, smoothness, fairness).
- **Ω (Observation Space):** What a robot can sense (partial, noisy).
- **O (Observation Function):** Maps true state to observations.
- **π (Policy):** Mapping from observations (and possibly state) to actions.

Key Properties

- **Partial Observability:** Robots do not have access to full world state.
- **Stochasticity:** Transitions and observations may be uncertain.

Controller Types

- **Reactive:** One observation $\rightarrow$ one action.
- **Receding Horizon / Planning:** Plans a sequence of future actions.

Taxonomy Axes

- **Centralized:** One controller for all robots.
- **Decentralized:** Each robot acts independently.
- **Distributed:** Robots act independently but share information.
- **Fully Observable vs Partially Observable**
- **Communicative vs Non-Communicative**
- **Cooperative vs Non-Cooperative**

Why This Matters

Taxonomy allows comparison of MRN methods across assumptions, scalability, safety, and realism. Week 1 focuses on understanding these assumptions, not algorithm implementation.